

TEAM DATA & BASIC CALCULATIONS

Names: _____ Team: _____

Record what happened

Workers: 4 Trial duration: 20 minutes

Completed bags at final location, B: _____

Approximate one-way carrying distance: _____ ft

Shovels used: _____ Other moving equipment: _____

Was the full trial supplied with empty bags and sand? Yes / No

If not, when and what ran out? _____

Weather / ground / route conditions: _____

Closure / fill standard used: _____

Jobs at the start and any changes: _____

Main bottleneck and evidence (waiting, queues, crowded area): _____

1. Experimental team production rate

$R = B \times 3 = _____ \times 3 = _____ \text{ bags per team-hour}$

This means one team of four at the conditions you measured.

2. Experimental average per person

$R \div 4 = _____ \div 4 = _____ \text{ bags per person-hour}$

This is an average within the team, not a promise that a solo worker can do every job at this rate.

3. Fill represented by the completed bags

$B \div 75 = _____ \div 75 = _____ \text{ cubic yards of fill (estimate)}$

No weighing. We use the same 75 bags $\approx 1 \text{ yd}^3$ assumption throughout.

4. Sustained planning rates for Part 2

Normal: $S = R \times 0.75 = _____ \text{ bags per team-hour}$

Difficult conditions: $S = R \times 0.50 = _____ \text{ bags per team-hour}$

Keep rates to one decimal place, or retain calculator precision.

5. What would improve production?

One change and the resource it needs: _____

Why adding people alone might not help: _____

Data-quality note

Circle: Full trial / Supply-limited / Weather-stopped / Other

If invalid or zero, teacher assigns a labeled practice rate. Do not invent measured results.